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RESEARCH ARTICLE

The Response of Soil Preferential Flow Characteristics to Vegetation Restoration in the Hilly and Gully Region of the Loess Plateau

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ABSTRACT

Preferential flow considerably affects soil infiltration in the hilly and gully regions of the Loess Plateau, where large-scale vegetation restoration under the Grain-for-Green Program has substantially altered the soil structure. However, the mechanisms underlying the vegetation restoration-mediated modification of preferential flow characteristics and the preferential flow-mediated internal regulation of infiltration partitioning remain insufficiently understood. In this study, we focused on forestlands and grasslands in the Yanhe River Basin, employing the double-ring infiltration method and dye-tracing technique to quantify the preferential flow characteristics and their contribution to total infiltration. The results revealed that forestlands exhibited the maximum dye depth (59.33 cm) and highest dye coverage (65.90%). The preferential flow ratios and length indices of forestlands (13.94% and 122.54%, respectively) and grasslands (25.42% and 120.03%, respectively) exceeded those of farmlands (9.53% and 100.01%, respectively). Despite relatively deeper and more continuous preferential pathways in forestlands, grasslands exhibited the highest contribution of preferential flow to total infiltration (46.79%), followed by those in forestlands (21.50%) and farmlands (23.05%); these results indicate that piston flow dominates the infiltration, with the preferential flow serving as a supplementary process. Enhanced preferential flow under vegetation restoration was associated with higher total porosity (54.44%–55.39%), lower bulk density (1.30–1.31 $g \cdot cm^{-3}$), and greater root biomass (105.56–170.22 $g \cdot m^{-2}$), whereas farmland exhibited a more compact soil structure with lower porosity (49.86%), higher bulk density (1.34 $g \cdot cm^{-3}$), and limited root development (18.45 $g \cdot m^{-2}$). Preferential flow acts as an internal regulator, redistributing infiltration pathways rather than directly controlling the total infiltration. In practice, grassland is recommended as the vegetation type for restoration on the Loess Plateau, considering the improvement in soil structure and the balancing effect on deep soil moisture.

1 | Introduction

Vegetation restoration has emerged as an effective land management strategy for mitigating land degradation on a global scale by improving soil structure and fertility, enhancing hydrological functions, and strengthening ecosystem services across degraded landscapes (Lal 2020; Sun et al. 2017). Large-scale restoration programs in degraded plateaus and drylands, such as afforestation and grassland revegetation in the Mediterranean region, African Sahel, and Chinese Loess Plateau, have significantly altered soil physical properties and infiltration patterns (Ilstedt et al. 2016; Berdugo et al. 2020; Liao et al. 2023). Although numerous studies have reported increased infiltration capacity and reduced surface runoff following vegetation restoration (Pan et al. 2023; Qiu et al. 2024), growing evidence suggests that such hydrological improvements are not solely governed by matrix (piston) flow (Beven and Germann 1982; Bouma 1981; Lin and Zhou 2008; Li et al. 2025) but are strongly mediated by preferential flow processes (Alaoui 2015; Bargués Tobella et al. 2014). However, the role of preferential flow in supporting or constraining the hydrological benefits of vegetation restoration remains poorly understood, particularly in thick loess deposits and plateau landscapes.

Preferential flow is a widespread hydrological process controlling rapid water and solute transport, primarily driven by macropore networks formed through biological activity, soil structural development, and land-use-induced porosity changes (Allaire et al. 2009; Alaoui 2015; Filipović et al. 2020; Latorre and Moret-Fernández 2022; Chu et al. 2025; Villamizar and Brown 2017). In the Loess Plateau of China, which has witnessed large-scale vegetation restoration-mediated conversion of farmland to forestland and grassland, the physicochemical properties and infiltration patterns of the soil have also changed accordingly (Liao et al. 2023; Qiu et al. 2024). Numerous researchers have investigated differences in the fundamental physical and chemical properties of soil resulting from various vegetation restoration approaches. For example, Wu et al. (2023) conducted a comprehensive investigation on the effects of different vegetation restoration practices on soil properties, including soil organic carbon content, bulk density (BD), total porosity (TP), saturated hydraulic conductivity (*Ks*), water retention curve, aggregate stability, and particle size distribution, in a semi-arid region of the Loess Plateau. Pan et al. (2023) quantitatively characterized the impact of vegetation restoration on *Ks* in the root zone, demonstrating that vegetation restoration significantly enhances *Ks*. Furthermore, Liu et al. (2025) reported that natural secondary forests exhibit lower soil volumetric water content, higher TP, and greater sand content than those in the artificial forests of *Pinus tabulaeformis*, *Thuja orientalis*, and *Robinia pseudoacacia*. With respect to soil chemical properties, Yuan et al. (2025) focused on the effects of grasslands and legumes on soil organic carbon sources following the conversion of farmland to grassland. Dong et al. (2024) examined the response of iron-bound organic carbon in the soil carbon pool to changes in vegetation recovery. Overall, large-scale vegetation restoration projects have considerably improved the fundamental physicochemical properties of soil.

Alterations in soil physicochemical properties are critical factors that impact preferential flow and infiltration. However, the spatial heterogeneity of the soil poses substantial challenges when investigating the mechanisms underlying preferential flow and infiltration. Previous studies have demonstrated that alterations in soil organic matter (SOM), BD, porosity, soil texture, and other parameters significantly impact preferential flow processes (Liao et al. 2023; Zhao et al. 2024). The macropores formed by plant roots notably impact preferential flow infiltration in the soil. The large-scale conversion of farmland to forest and grassland has enriched root systems within the soil. Plant roots not only significantly increase the number of macropores in the soil by forming root channels but also contribute to humus formation during decomposition (Levia and Frost 2003; Schwärzel et al. 2012), promoting the development of soil aggregates and macropores, thereby establishing preferential flow infiltration pathways in the soil (Zhang et al. 2025). Guan et al. (2025) conducted further research on the effects of roots of different diameters on preferential pathways, concluding that roots of all diameters promote the formation of preferential flow pathways, with fine roots exerting a more pronounced effect than observed for coarse roots. Similarly, Wen et al. (2025) investigated the relationship between root system density characteristics and soil infiltration capacity and preferential flow; increases in root length density, root surface density, root volume density, and root weight density enhanced soil infiltration capacity and preferential flow, and concluded that root system characteristics are the dominant factors influencing soil preferential flow. Additionally, the initial soil moisture content affects preferential flow by altering the soil pore network (Greve et al. 2012). The research by Hou et al. (2023) indicates that *I*- and *Λ*-shaped soil cracks also facilitate the generation of preferential flow. In summary, the formation and development of preferential flow in soil are strongly impacted by factors such as soil physicochemical properties, plant root characteristics, and soil structure.

Despite extensive research on vegetation restoration and soil hydrological responses in the Loess Plateau, the underlying key mechanisms remain unclear. Previous studies have largely focused on preferential flow characteristics or infiltration responses under individual land use types, often treating preferential flow as a descriptive outcome rather than an internal regulator of the overall infiltration process (Luo et al. 2008; Bargués Tobella et al. 2014; Zhang et al. 2019; Qiu et al. 2023). In particular, it remains uncertain whether forestland and grassland revegetation promote preferential flow through similar or contrasting root system-driven mechanisms of soil structural development and to what extent shifts in preferential flow are directly attributable to land-use type versus indirectly mediated by soil physical properties and root-induced pore networks (Cai et al. 2024; Zhao et al. 2024). Addressing these gaps is essential to advance the process-based understanding of how vegetation restoration reshapes hydrological functioning in degraded plateau ecosystems.

Therefore, this study investigated how vegetation restoration reshapes infiltration processes on the Loess Plateau. Specifically, we aimed to (i) quantify how vegetation restoration modifies infiltration pathways and preferential flow characteristics across farmland, forestland, and grassland; (ii) compare the contribution of preferential flow (CPF) to infiltration; and (iii) disentangle the direct and indirect effects of land-use change on preferential flow characteristics by integrating soil physical properties and root biomass using partial least squares (PLS) structural equation modeling (PLS-SEM). To achieve these objectives, three replicate plots were established for each land-use type in the Loess Plateau hilly gully region. Field experiments combining double-ring infiltration measurements with dye tracer staining were conducted to quantify infiltration dynamics and visualize preferential flow patterns across land uses. This integrated approach provides a process-based foundation for elucidating and predicting hydrological functional changes induced by vegetation restoration.

2 | Materials and Methods

2.1 | Study Area

The study area is located in the National Field Scientific Observation and Research Station of the Ansai Farmland Ecosystem, Shaanxi Province (36°30′45″—37°19′03″ N, 108°05′44″—109°26′18″ E). It lies within the mountainous terrain of the inland Loess Plateau hilly-gully area (Figure 1) and is characterized by elongated gullies and an ecologically fragile environment. The climate is mid-temperate, continental, and semi-arid monsoonal, characterized by limited annual precipitation. The rainy season occurs from July to September, which aligns with the region of the "Three North Protective Forest Project." The mean annual precipitation is 505 mm, and the mean annual temperature is 8.8°C. The dominant soil type was loess-derived silt loam, which originated from the parent material. This soil is distinguished by its homogeneity, porosity, loose structure, and yellow-to-light brown color.

The preliminary phase of this study was conducted from September to November 2024. The farmland selected by the research institute is representative of primary crops, namely potatoes and broomcorn millet, both of which have been harvested. *Robinia pseudoacacia* characterizes the forestland, whereas the grassland is represented by *Phragmites* and *Artemisia capillaris*. Three experiment sites were selected for each land-use type to conduct dye tracing experiments aimed at investigating the preferential flow characteristics across different land uses. A total of nine experimental sites were selected, with three vertical soil profiles excavated at each dye tracing sample site. Basic information on the experiment sites is summarized in Table 1.

2.2 | Test Procedure and Sample Collection

2.2.1 | Test Procedure

With the increasing understanding of preferential flow infiltration, a wide range of research methods has been developed, such as the staining tracer method (Ghodrati and Jury 1990; Kasteel et al. 2002; Nobles et al. 2010), the CT scanning method (Luo et al. 2008; Lissy et al. 2020; Gackiewicz et al. 2022), the penetration curve method (Kamra and Lennartz 2005; Koestel et al. 2011; Latorre and Moret-Fernández 2022), and the ground-penetrating radar method (Di Prima et al. 2022). Among the available methods, the staining tracer method using Brilliant Blue solution is the most widely adopted in field experiments owing to its simplicity and practicality under field conditions (Flury and Flühler 1995; Zhang et al. 2025). Therefore, the double-ring infiltration method was employed for dye tracing. To ensure optimal conditions, a precipitation-free period was required within one week before and after the experiment. After nine sample plots across the three land-use types were identified through field surveys, a double-ring infiltration experiment combined with dye tracing was conducted. The

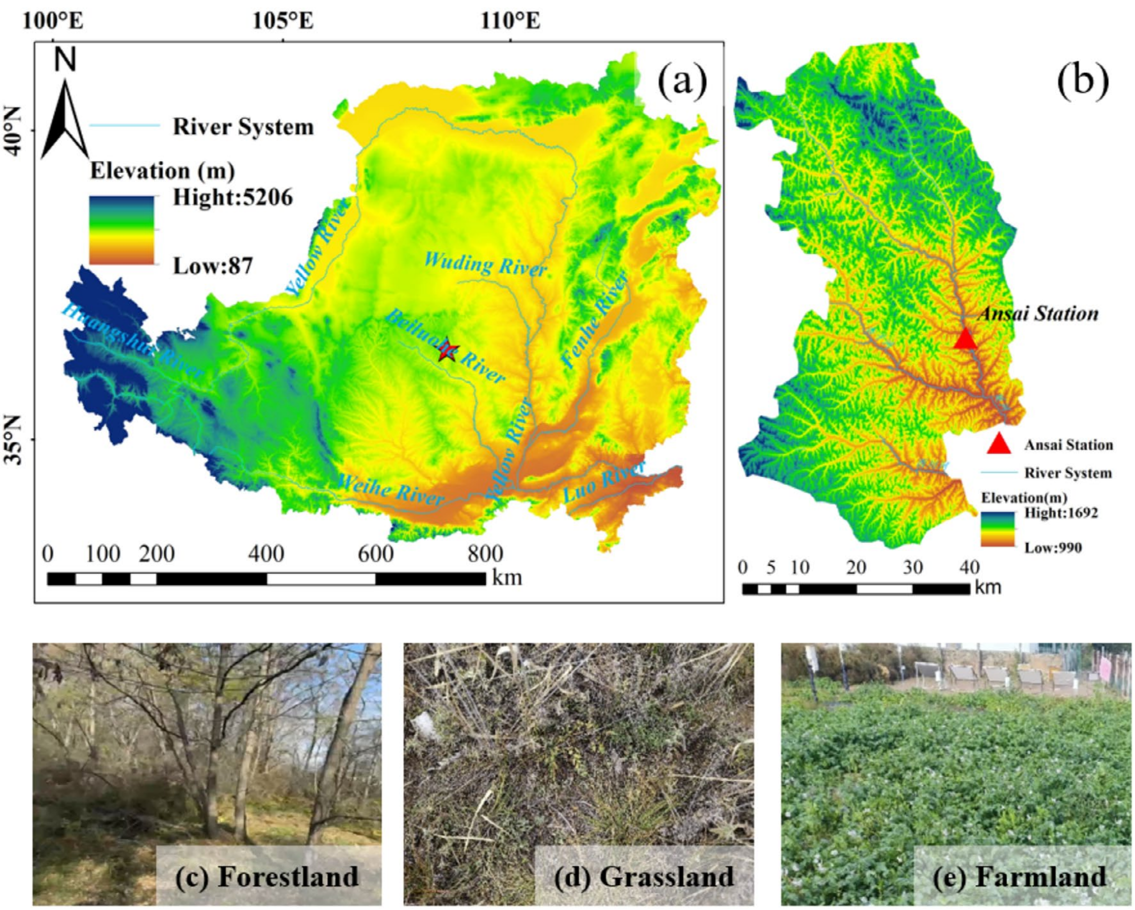

FIGURE 1 | General situation of the study area and the actual map of test plot. [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 | Basic information about the experiment sites.

Land use type	Vegetation	Location	Elevation/m	Soil texture	Age of stand/year	Plant height/m	Vegetation coverage/%
Forestland 1	*Robinia pseudoacacia*	109°19′17″ E,36°51′21″ N	1229	Silt loam	20	12	75
Forestland 2	*Robinia pseudoacacia*	109°19′04″ E,36°51′08″ N	1275	Silt loam	20	11	90
Forestland 3	*Robinia pseudoacacia*	109°19′17″ E,36°51′21″ N	1229	Silt loam	20	14	80
Grassland 1	*Phragmites*	109°19′17″ E,36°51′22″ N	1225	Silt loam	20	1.1	85
Grassland 2	*Phragmites*	109°19′11″ E,36°51′14″ N	1277	Silt loam	20	0.9	70
Grassland 3	*Artemisia capillaries*	109°19′12″ E,36°51′16″ N	1274	Silt loam	20	0.2	65
Farmland 1	*Broomcorn millet*	109°19′19″ E,36°51′23″ N	1217	Silt loam	30	0.7	10
Farmland 2	*Potato*	109°19′18″ E,36°51′23″ N	1221	Silt loam	30	0.3	15
Farmland 3	*Broomcorn millet*	109°19′18″ E,36°51′25″ N	1209	Silt loam	30	0.8	13

specific procedures used are described below. The materials required for the experiment included scissors and a double-ring infiltration apparatus (a double-ring apparatus composed of two concentrically arranged square iron rings). The inner ring had a side length of 30 cm, and the outer ring had a side length of 60 cm; both rings were 8.0 cm in height, water buckets, shovel, cardboard, Brilliant Blue solution, water, ring knives, aluminum boxes, sampling bags, smartphone camera, measuring tape, soil auger, plastic film, ruler, and scale. Subsequently, the surface layer of dead leaves and fallen branches, 0–5 cm deep, was carefully removed to expose the underlying soil substrate. The use of stick rulers on the inner side of both the inner and outer rings facilitated the interpretation of the water level data. For the installation of the double-ring infiltration meter proposed by Zhang et al. (2019), a double ring was inserted 1.0 cm into the ground surface and carefully leveled. To prevent lateral seepage, enough soil should be compacted around the outer sides of both the outer and inner rings. During installation, it is crucial to maintain a level position and minimize disturbances to the ground surface. To reduce the topsoil loss during the watering process, a piece of cardboard was placed on the soil surface. The water flow can then be directed onto the cardboard, which acts as a cushion to reduce the impact force on the water. A Brilliant Blue solution was prepared at a concentration of $4.0\,g \cdot L^{-1}$ and then introduced into the inner ring. Subsequently, an equal volume of clean water was added to the outer ring. At the beginning of the experiment, both the inner and outer rings were filled simultaneously with the same water level, with an initial depth of 4.0 cm. The experiment was initiated by pouring water from a bucket. The stopwatch was started simultaneously with the application of water, and the time (in minutes) required for the water level in the inner ring to decrease by 0.5 cm is recorded. When the water level in the inner ring had decreased to 1.0 cm, sufficient water was added to increase the water level to 4.0 cm. The water levels were maintained relatively consistently throughout the infiltration process in both the inner and outer rings, ensuring that the infiltration followed a vertical one-dimensional pattern. Stable infiltration was achieved when three consecutive measurements yielded approximately equal time intervals (Zhang et al. 2019). At this point, the addition of water and solution was discontinued. Once no surface water accumulation was visible on the soil surface, the double-ring infiltration area was covered with a plastic film to protect it from precipitation and disturbances by wildlife. Following a 24-h duration, the plastic film and metal double ring were removed. As shown in Figure 2a, three vertical sections were excavated along the inner frame edge at distances of 5.0, 15.0, and 25 cm from the center of the frame. Each section was 30 cm wide and extended downward by approximately 20 cm beyond the maximum dyeing depth. The soil sections were carefully trimmed, and their edges were marked using a ruler. Finally, the images of each section were captured using an iPhone 14 Pro Max.

2.2.2 | Soil Sample Collection and Index Testing

After excavating each pit, soil samples were collected from the unstained profile from top to bottom using a ring knife ($100\,cm^3$) and self-sealing bags. Ring knife samples were collected at depths of 20, 50, 80, and 110 cm. To ensure complete capture of stained profiles of soil preferential flow under different land-use types and to avoid staining depths exceeding the sampling depth, sampling strategies were adjusted according to vegetation type and the expected depth of preferential flow. For

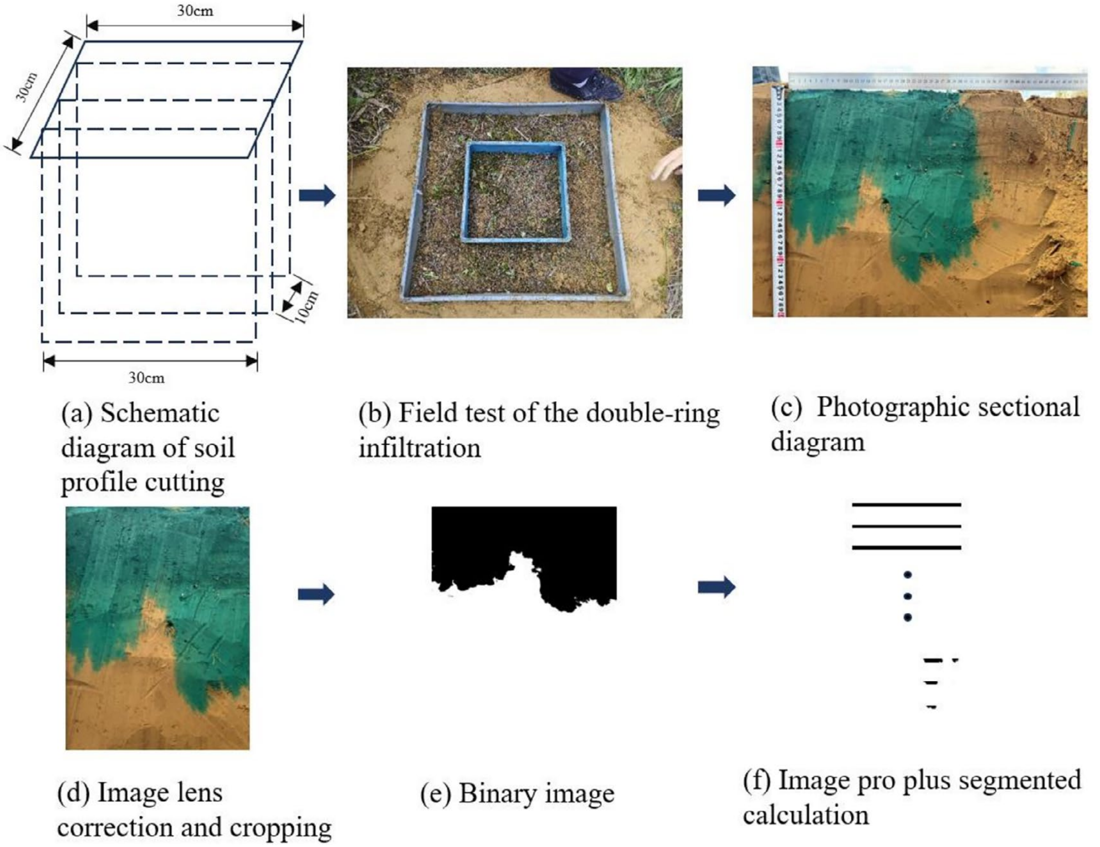

FIGURE 2 | Dyeing image digitization process. [Colour figure can be viewed at wileyonlinelibrary.com]

forestlands, where relatively deeper preferential flow was anticipated, soil samples were collected at 20 cm intervals down to a depth of 200 cm using self-sealing bags. For grasslands and farmlands, sampling was conducted at a depth of 160 cm. Before the experiment, soil samples were collected from locations adjacent to the experimental pits using a soil auger to determine the initial soil moisture content. Additionally, three soil samples measuring 5.0 cm × 5.0 cm × 10 cm were collected from the inner ring surface layer using perforated net bags to calculate the root biomass (R-bio) within the soil profile. All soil samples and ring knives were transported to the laboratory for basic physical and chemical analyses. Soil BD was measured using the oven-drying method at 105°C until a constant weight was achieved. The parameters TP, field capacity (FC), capillary moisture capacity (CMC), capillary porosity (CP), non-CP (N-CP), and *Ks* were analyzed using the standard ring-knife method (Bao 2000). SOM content was determined by potassium dichromate oxidation using an external heating method (Dong et al. 2022). The particle size distribution of the soil was analyzed using a Mastersizer 2000 laser particle size analyzer.

2.3 | Data Analyses

2.3.1 | Preferential Flow Characteristic Parameters and Contribution

Figure 2 illustrates the image-processing workflow for staining analysis. The soil profile stained images were processed using Photoshop 2023 for lens correction, cropping, and noise reduction. Subsequently, the images were converted into black-and-white binary images by selecting an appropriate color range and adjusting the threshold levels. Stained areas were assigned a pixel value of 0 (black), whereas unstained areas were assigned a pixel value of 255 (white). Three black-and-white binary images from each sampling site were superimposed, and Image-Pro Plus v6.0 software was employed to calculate the dye coverage. Finally, we calculated various preferential flow indicators, including dye coverage (DC), uniform flow depth (U_F), preferential flow ratio (P_{FR}), and length index (LI). Additionally, we determined the preferential infiltration volume (PIV) and its contribution to the total infiltration volume (TIV). The specific equations used for calculating these indicators are provided below.

DC: Ratio of the dyed area to the total area of the soil profile. This indicator provides an intuitive representation of the overall extent of dye penetration into the soil (Flury and Flühler 1995) and can be calculated using Equation (1) as follows:

$$DC = \left(\frac{D}{D + ND}\right) \times 100\% \tag{1}$$

where D is the dyed area of the soil profile (cm^2), and ND is the non-dyed area of the soil profile (cm^2).

U_F: Soil depth at which the stained area ratio reaches at least 80% within the vertical soil profile. Once this threshold is exceeded, the infiltration process is predominantly characterized by piston flow infiltration, indicating preferential flow (Flury and Flühler 1995).

P_{FR}: Percentage of the total stained area occupied by the preferential flow area in the vertical soil profile. It represents the ratio of the stained area below the piston flow to the total stained area. This value indicates the extent of preferential flow development (Bargués Tobella et al. 2014) and can be expressed using Equation (2) as follows:

$$P_{FR} = \left(1 - \frac{U_F \times W}{D}\right) \times 100\% \tag{2}$$

where W is the horizontal width of the soil profile (30 cm), and D is the total stained area of the vertical soil profile (cm^2).

LI: Sum of the absolute values of the differences in the ratio of stained areas between adjacent soil layers at a given depth, divided by the number of equal segments into which the vertical profile depth was divided (van Schaik 2009). A higher LI value indicates greater spatial heterogeneity in preferential flow. LI can be calculated using Equation (3) as follows:

$$LI = \sum_{\mathrm{i}=1}^{\mathrm{n}} \left|DC_{\mathrm{i}+1} - DC_{\mathrm{i}}\right| \tag{3}$$

where DC_{i+1} and DC_i represent the DC (%) of the $(i+1)$th and i-th layers in the vertical soil profile, respectively, and n is the total number of soil layers.

PIV: It denotes the volume of water that infiltrates through preferential flow pathways during a specific field test (Mei et al. 2018) and can be expressed using Equation (4) as follows:

$$PIV = \sum_{\mathrm{i}=0}^{\mathrm{i}=ID} V_{\mathrm{i}} - \sum_{\mathrm{j}=0}^{\mathrm{j}=U_F} V_{\mathrm{j}} \tag{4}$$

where ID denotes the infiltration depth (cm), and V_i and V_j represent the infiltration water volumes of the i-th and j-th soil layers, respectively. These variables can be expressed using Equation (5) as follows:

$$V = 10 \sum_{\mathrm{u}=1}^{\mathrm{u}=4} \left(\theta_{\mathrm{tu}} - \theta_{\mathrm{iu}}\right) \times d \times v_s / 80 \tag{5}$$

where θ_{tu} is the soil moisture content after the infiltration test (%); θ_{iu} is the initial soil moisture content (%); and u is the soil sample sequence with a maximum value of 4, as soil samples were collected at 20 cm intervals within each 80 cm deep soil profile in this study. d is the BD ($g \cdot cm^{-3}$) of the corresponding soil layer, and V_s is the volume of the soil sample (cm^3).

CPF: This parameter quantifies the proportion of total infiltration attributable to preferential flow infiltration and can be expressed using Equation (6) as follows:

$$CPF = \frac{PIV}{TIV} \times 100\% \tag{6}$$

2.3.2 | Statistical Analyses

In this study, one-way analysis of variance (ANOVA) was performed to assess the differences in the fundamental physical and chemical characteristics of the soil and its preferential flow characteristics (DC, U_F, P_{FR}, and LI). Pearson's correlation analysis was used to explore the relationships between these basic soil properties and the preferential flow characteristics. Additionally, PLS-SEM was applied to quantify the direct and indirect effects of land use type, soil properties, and root biomass on preferential flow characteristics. The model was estimated using a PLS path modeling algorithm implemented in the R package plspm. Furthermore, during the construction of the PLS-SEM model, the preferential flow characteristics were separated into morphological and functional components. Morphological features describe the spatial organization of the preferential flow paths and were quantified using DC and LI. Functional contribution reflects the hydrological dominance of preferential flow and was assessed using U_F and P_{FR}. One-way ANOVA and Pearson's correlation analysis were performed using SPSS Statistics (version 27.0; SPSS Inc., Chicago, IL, USA). PLS-SEM was performed using R version 4.5.0.

3 | Results

3.1 | Response of Soil Physical and Chemical Properties to Vegetation Restoration

As shown in Table 2, soil BD and organic matter content exhibited significant responses to changes in land-use type ($p < 0.05$). The conversion of farmland to forest and grassland led to a decrease in soil BD. Specifically, the BD of farmland was considerably higher than that of forestland, at a ratio of 1.03. In contrast, no significant differences were observed between forestlands and grasslands in terms of BD. SOM content was higher in forestland and grassland areas than in farmland areas. In forestland areas, the organic matter content was significantly greater than that in farmlands, with a ratio of 1.69, and exhibited no significant differences compared with that in grasslands. Both forestland and grassland exhibited higher levels of sand content, TP, CP, FC, and CMC than those in farmland, with increases of 11.11 and 4.56; 9.19 and 11.09; 9.66 and 12.34; 17.92 and 27.86; and 13.18% and 15.29%, respectively. These findings demonstrated that forestland use resulted in the most substantial improvements in soil physical and chemical properties, further supporting the effectiveness of the Grain-for-Green Program in enhancing soil quality.

3.2 | Soil Preferential Flow Characteristics in Farmland, Forestland, and Grassland

Figure 3 presents the binary images of the excavated stained soil profiles for different land-use types after image capture and post-processing. Overall, grasslands displayed a more pronounced preferential flow than forestland and farmland soils, which exhibited a weaker preferential flow. The stained areas within each soil profile decreased with increasing depth, indicating preferential flow. However, the extent of this reduction varied across land-use types. Upon individual examination of each profile, forest land was found to have the largest stained area, followed by grassland and farmland. The soil profiles of Grasslands 1 and 2, Forestland 2, and Farmland 2 exhibited significant heterogeneity in their staining patterns, whereas the remaining profiles exhibited relatively more uniform staining.

A DC analysis was conducted at 1 cm intervals for each cross-section. Figure 4 presents the distribution curve of DC with soil

TABLE 2 | The basic physical and chemical properties of soils under different vegetation types.

Land type	Clay (%)	Silt (%)	Sand (%)	BD ($g \cdot cm^{-3}$)	TP (%)	CP (%)	N-CP (%)	FC (%)	CMC (%)	KS ($m \cdot h^{-1}$)	SOM ($g \cdot kg^{-1}$)	R-bio ($g \cdot m^{-2}$)
Forestland	9.72 ± 1.30a	15.90 ± 1.30a	74.38 ± 2.50a	1.30 ± 0.03b	54.44 ± 2.37a	50.75 ± 1.98a	3.69 ± 0.74a	23.62 ± 2.09a	39.08 ± 2.11a	23.24 ± 19.21a	3.83 ± 0.20a	105.56 ± 33.64a
Grassland	11.95 ± 1.81a	18.06 ± 1.65a	69.99 ± 3.47b	1.31 ± 0.02a	55.39 ± 4.23a	51.99 ± 4.08a	3.40 ± 0.16a	25.61 ± 2.20a	39.81 ± 3.45a	7.51 ± 2.33a	3.42 ± 0.79ab	170.22 ± 143.26a
Farmland	12.78 ± 1.81a	20.28 ± 3.76a	66.94 ± 5.47a	1.34 ± 0.01a	49.86 ± 1.97a	46.28 ± 2.33a	3.58 ± 0.41a	20.03 ± 3.63a	34.53 ± 1.81a	13.10 ± 4.35a	2.26 ± 0.68b	18.45 ± 8.49b

Note: The data in the table are presented as means ± standard deviation. Different letters within the same column denote statistically significant differences among land use types ($p < 0.05$).

depth. The decrease in DC across different land-use types with increasing soil depth suggests an uneven distribution of water flow or varying degrees of preferential flow. Forestland exhibited the highest maximum dyeing depth (59.33 cm) and DC (65.9%), surpassing grassland (49.33 cm and 48.07%, respectively) and farmland (32.00 cm and 34.4%, respectively). Specifically, the maximum infiltration depth in forestland was 1.2 times greater than that in grassland and 1.85 times greater than that in farmland. Similarly, the DC in forestland was 1.37 times higher than in grassland and 1.92 times higher than in farmland. The differences in DC between farmlands and forestlands were significant; however, no differences were observed between grasslands and farmlands. The DC in farmland and forestland decreased rapidly within the span of a few centimeters. In contrast, the DC distribution curves for Forestland 2 and Grassland display a "big wave" pattern, characterized by a relatively gradual decrease. However, the decrease in Grassland 3 was notably steep. Figure 4 also indicates that, in the sample plots representing different land-use types selected for this study, DC in the surface layer (0 cm to 20 cm or 30 cm) was nearly 100% across all soil profiles. Additionally, in Forestland 1 and Forestland 3, an increase in DC was observed at depths of up to 40 cm.

Through further analysis of black-and-white binary images of soil staining profiles across different land-use types, three indicators of preferential flow characteristics were identified: U_F, P_{FR}, and LI. Figure 5 presents the specific preferential flow indicators for each land-use type. First, the U_F values estimated for forestland, grassland, and farmland were 45.58, 28.24, and 24.6 cm, respectively. The infiltration depth in forestland was the highest, exceeding that of grassland by 61.4% and that of farmland by 85.28%. The P_{FR} followed the order: grassland (25.42%) > forestland (13.94%) > farmland (9.53%). The LI of forestland (122.54%) was 1.02 times that of grassland (120.03%) and 1.23 times that of farmland (100.01%). The length indices of forestland and grassland areas were similar, and the three farmland sample plots all exhibited an LI of 100%. However, no notable variations were observed in U_F, P_{FR}, or LI between forestlands, grasslands and farmlands.

3.3 | Contribution of Preferential Flow to Infiltration

Figure 6 illustrates the specific distribution of the preferential flow infiltration. Overall, piston flow was the dominant mechanism for all land-use types, with preferential flow serving as a supplementary pathway. Preferential flow infiltration accounted for 1.54% to 59.13% of total infiltration in forestlands, with an average contribution of 21.50%. In grasslands, it ranged from 8.41% to 71%, with an average of 46.79%, and varied between 2.75% and 57.99%, with an average of 23.05%. In Forestland 1, preferential flow infiltration was primarily observed in the 60–80 cm soil layer, contributing to 76.85% of the total infiltration in this layer. For Forestland 2, preferential flow infiltration occurred in the 20–40 cm and 40–60 cm layers, accounting for 94.70% and 100% of the corresponding total preferential flow infiltration, respectively. In Forestland 3, preferential flow infiltration was concentrated in the 40–60 cm layer, representing 27.11% of the total infiltration in this layer. In Grassland 1, preferential flow infiltration was distributed across the 20–40, 40–60, and 60–80 cm soil layers, with 80.68% occurring in the 20–40 cm layer and 100% in the other two

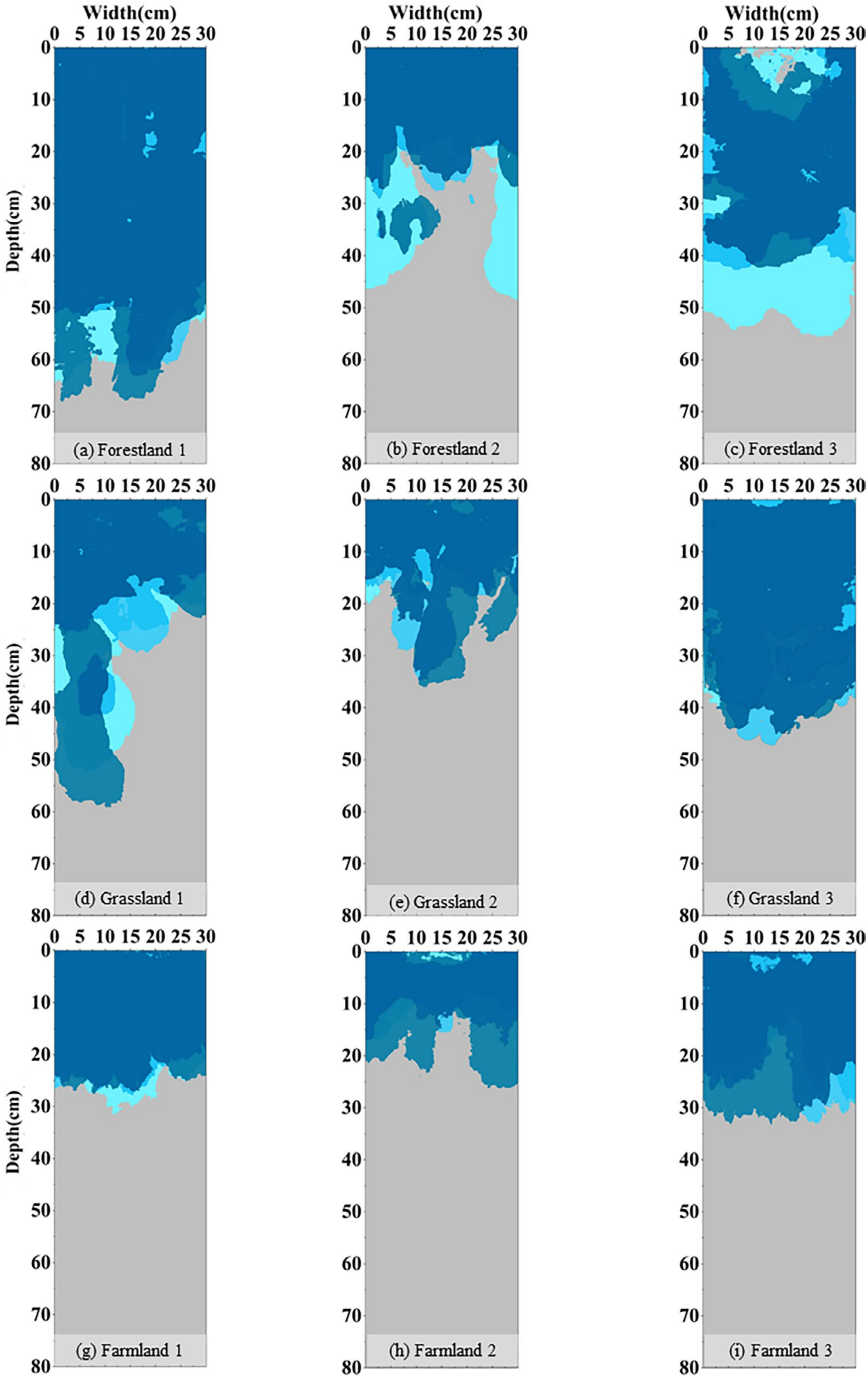

FIGURE 3 | Dyed superimposed images of different land use types. [Colour figure can be viewed at wileyonlinelibrary.com]

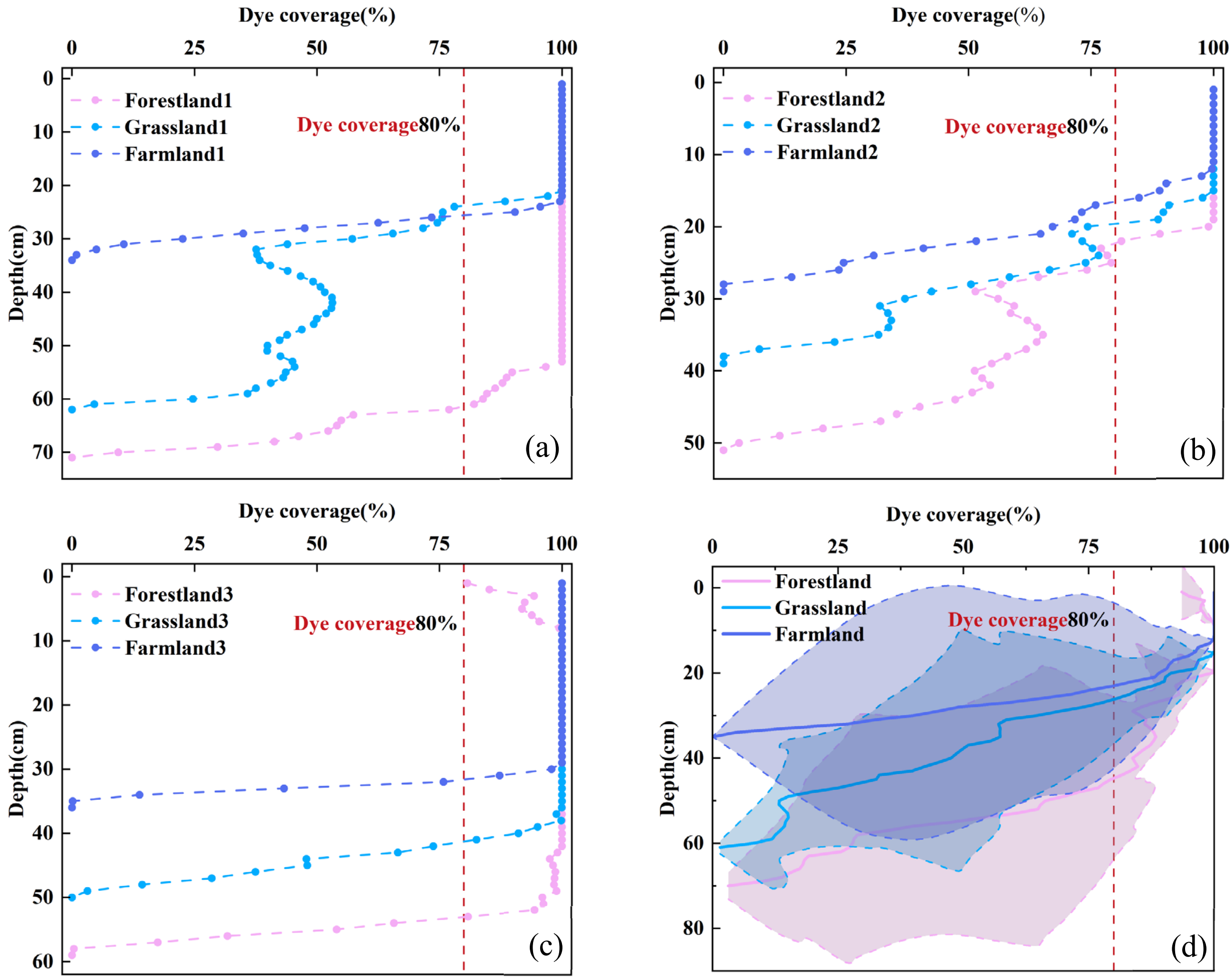

FIGURE 4 | The distribution curves illustrate the variation in dye coverage across different land use types with respect to soil depth. (a), (b), and (c) correspond to Plot No. 1, Plot No. 2, and Plot No. 3 respectively; (d) represents the variation curve of the average dye coverage across different land use types with soil depth. The shaded areas in panel d indicate the error ranges of dye coverage among the various land use types. [Colour figure can be viewed at wileyonlinelibrary.com]

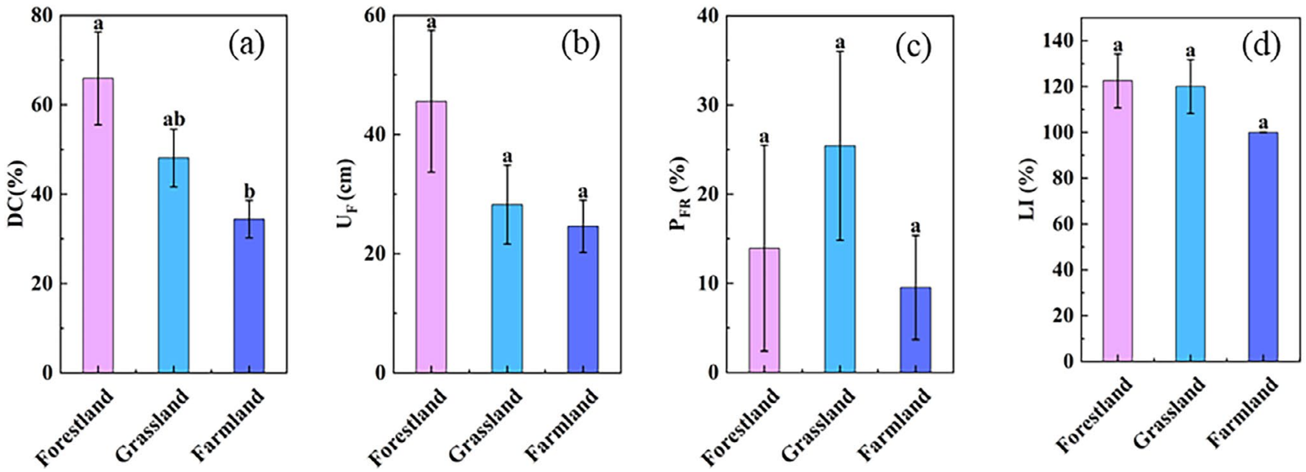

FIGURE 5 | A comparison chart of preferential flow indicates different land use types: (a) dye coverage, (b) uniform flow depth, (c) preferential flow ratio, and (d) length index. [Colour figure can be viewed at wileyonlinelibrary.com]

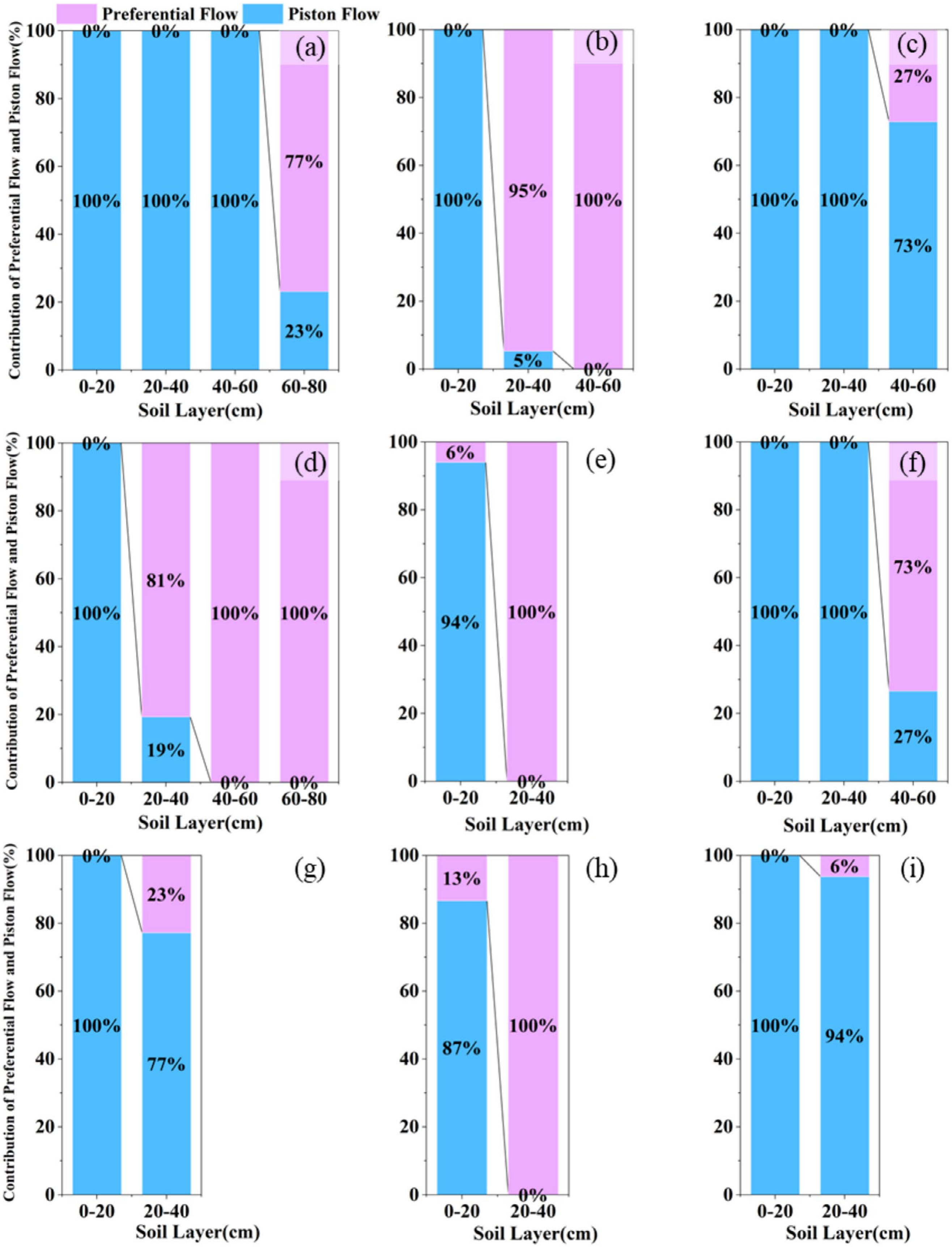

FIGURE 6 | The contribution ratio diagrams of preferential flow and piston flow for different land use types: (a) to (i) represent Forestland 1, Forestland 2, Forestland 3, Grassland 1, Grassland 2, Grassland 3, Farmland 1, Farmland 2, and Farmland 3 respectively. [Colour figure can be viewed at wileyonlinelibrary.com]

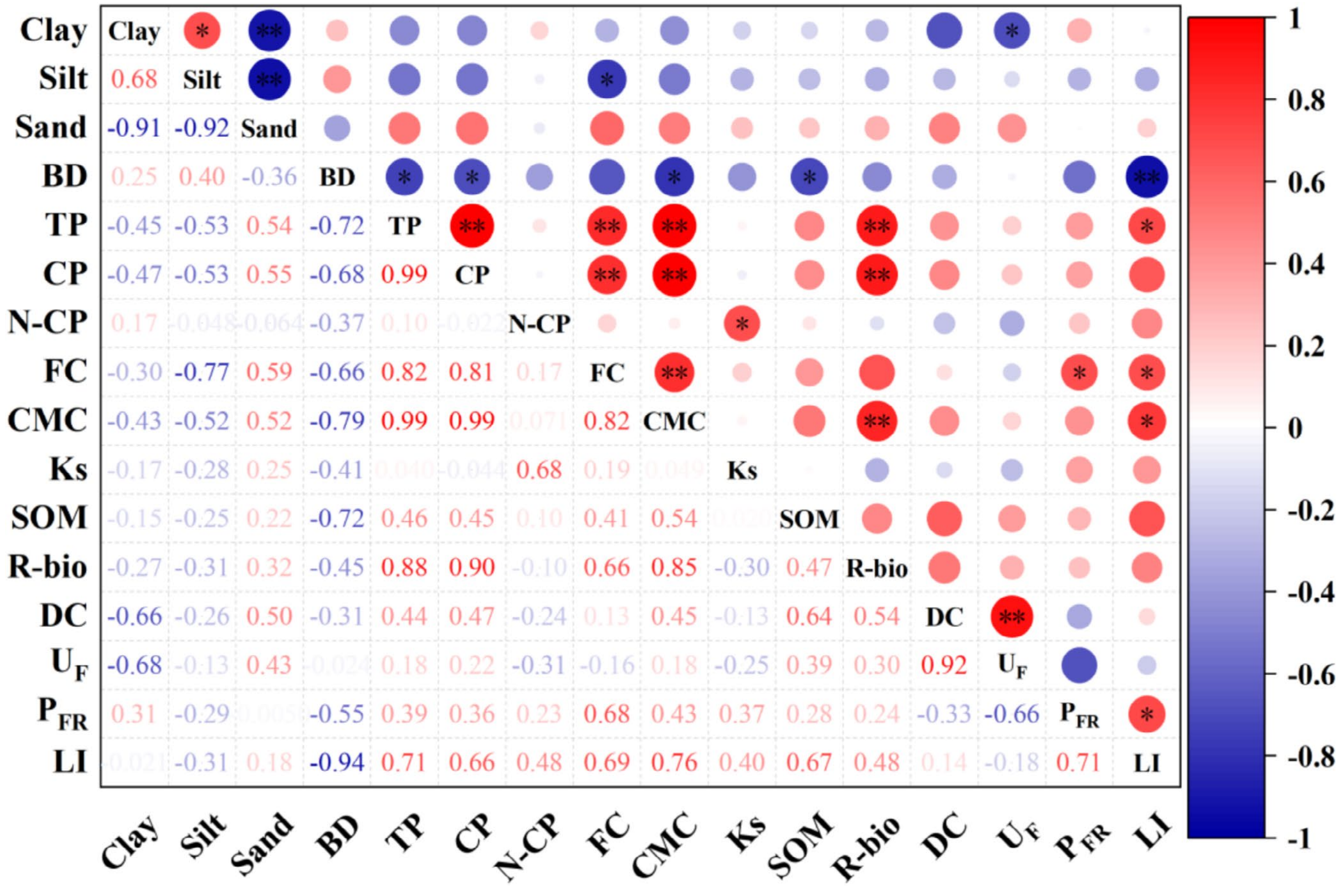

FIGURE 7 | Correlation analysis table of basic physical and chemical properties of soil and preferential flow characteristics. Red indicates positive correlation; blue indicates negative correlation, * indicates $p<0.05$, ** indicates $p<0.01$, and *** indicates $p<0.001$. BD: Soil bulk density; TP: Total soil porosity; CP: Soil capillary porosity; N-CP: Soil non-capillary porosity; FC: Field capacity; CMC: Capillary water holding capacity; Ks: Saturated hydraulic conductivity; SOM: Soil organic matter; R-bio: Root biomass; DC: Dye coverage; U_F: Uniform flow depth; P_{FR}: Preferential flow ratio; LI: Length index. [Colour figure can be viewed at wileyonlinelibrary.com]

layers, whereas in Grassland 2, preferential flow infiltration was distributed in the 0–20 cm and 20–40 cm soil layers, accounting for 5.96% and 100% of the infiltration in these layers, respectively. However, in Grassland 3, preferential flow infiltration occurred primarily in the 40–60 cm soil layer, representing 73.39% of the total infiltration in this layer. For Farmland 1, preferential flow infiltration was concentrated in the 20–40 cm layer, contributing 22.82% of the total infiltration. In Farmland 2, preferential flow infiltration occurred in both the 0–20 cm and 20–40 cm layers, accounting for 13.26% and 100% of the total infiltration in each layer, respectively, and in Farmland 3, preferential flow infiltration was distributed in the 20–40 cm layer and accounted for 6.18% of the total infiltration in this layer. Although piston flow constituted a relatively high proportion of the total flow, the proportion of preferential flow infiltration in grasslands was significantly higher than that in forestlands and farmlands.

3.4 | Factors Impacting Preferential Flow Characteristics

For a comprehensive analysis of the correlation between the fundamental soil physical and chemical properties and their preferential flow characteristics, please refer to Figure 7. The basic physical and chemical properties of the soil are significantly associated with its preferential flow characteristics. Specifically, the U_F exhibited a significant negative correlation with soil clay content ($p<0.05$), whereas the P_{FR} correlated significantly positively with FC ($p<0.05$). The LI exhibited significant positive correlations with TP, FC, and CMC ($p<0.05$) and an extremely significant negative correlation with BD ($p<0.01$). Furthermore, BD exhibited significant negative correlations with TP, CP, CMC, and SOM ($p<0.05$). TP demonstrated highly significant positive correlations with CP, FC, CMC, and R-bio ($p<0.01$). Similarly, CP exhibited extremely significant positive correlations with FC, CMC, and R-bio ($p<0.01$). N-CP correlated significantly positively with *Ks* ($p<0.05$), and CMC exhibited an extremely significant positive correlation with R-bio ($p<0.01$).

3.5 | Mechanistic Pathways Linking Vegetation Restoration, Preferential Flow, and Infiltration Partitioning

The PLS-SEM analysis revealed a hierarchical linkage between land use, root biomass, soil properties, preferential flow characteristics, and infiltration partitioning (Figure 8). Land use had a clear positive effect on the root biomass (path coefficient = 0.53). At the same time, land use showed a strong negative total effect on soil properties (total effect = −0.90), which

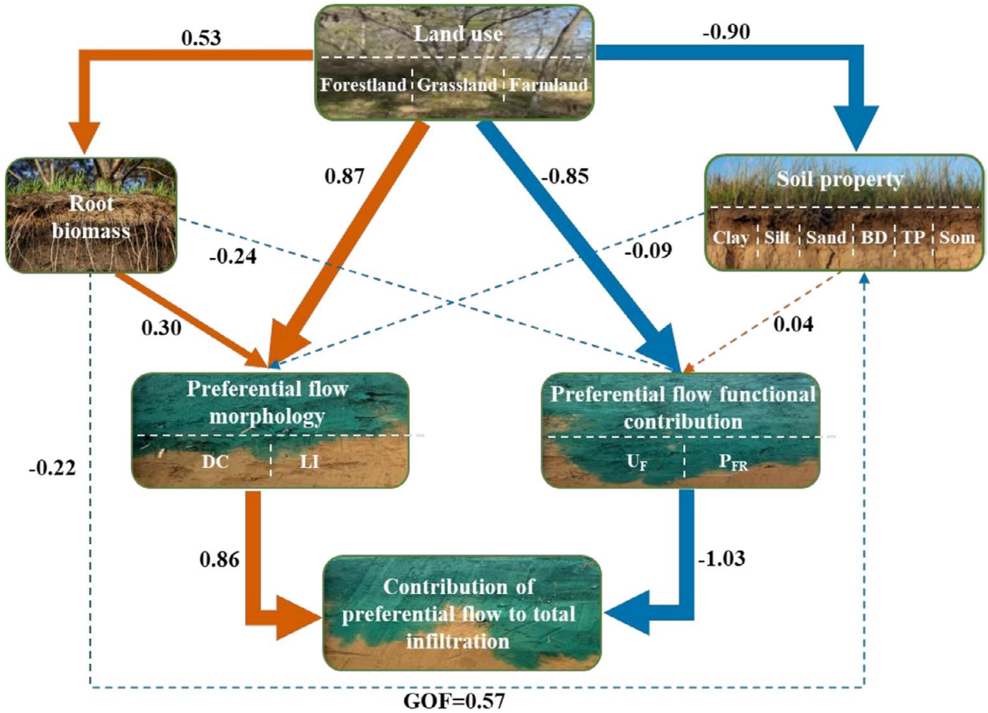

FIGURE 8 | Structural equation model illustrating the pathways linking land use, root biomass, soil properties, preferential flow morphology, preferential flow functional contribution, and the contribution of preferential flow to total infiltration. Solid arrows indicate relatively strong paths (standardized coefficients ≥ 0.25), whereas dashed arrows represent weaker paths (< 0.25). Arrow colors denote the direction of effects, with orange representing positive effects and blue representing negative effects. Line thickness corresponds to the magnitude of standardized path coefficients: 0 - |±0.25| (1 pt), |±0.25 - ±0.5| (4 pt), |±0.5 - ±0.75| (6 pt), and >|±0.75| (8 pt). Numbers along the arrows represent indicate standardized path coefficients. The model goodness of fit (GoF) is 0.57. [Colour figure can be viewed at wileyonlinelibrary.com]

was largely controlled by the direct pathway, indicating substantial vegetation-related modification of soil physical conditions. Preferential flow morphology was mainly impacted by land use, with a strong positive total effect (0.87) resulting from both direct and indirect pathways through root biomass and soil properties. In contrast, land use exerted a strong negative total effect on preferential flow functional contribution (−0.85). Root biomass contributed only weakly to the preferential flow functional contribution via indirect pathways. Soil properties had a moderately negative effect on preferential flow morphology (−0.24), whereas their effect on the preferential flow functional contribution was weak and positive (0.04). The preferential flow functional contribution showed a dominant impact on the contribution of the preferential flow to the total infiltration, with a strong negative path coefficient (–1.03). The preferential flow morphology has a secondary but positive effect (0.86). These results indicate that the preferential flow morphology and functional contribution respond differently to vegetation-driven changes in soil and root systems, with the functional efficiency of preferential flow playing a key role in regulating its contribution to total infiltration.

4 | Discussion

4.1 | Effects of Vegetation Restoration on Preferential Flow Characteristics

In this study, we evaluated the preferential flow characteristics of farmlands, forestlands, and grasslands in the hilly and gully regions of the Loess Plateau. The results indicate that the conversion of farmlands to forestlands and grasslands significantly enhances DC, P_{FR}, and LI (Figure 5). These findings align with those of previous studies, suggesting that long-term vegetation restoration measures generally improve the regional soil infiltration capacity and facilitate the formation and development of preferential flow in loess soils (Qiu et al. 2024; Xu et al. 2025). The correlation analysis between soil physico-chemical properties and preferential flow characteristics revealed that the U_F correlated significantly negatively with clay content; the P_{FR} correlated significantly positively with FC; and the LI correlated significantly positively with TP, FC, and CMC, while correlating significantly negatively with BD (Figure 7). This result is attributed to vegetation restoration, which changes the physical and chemical properties of the root systems and soil, thereby influencing the soil infiltration characteristics of the loess and promoting preferential flow (Xu et al. 2025). Similar responses have been documented in restored ecosystems across the Loess Plateau (Mei et al. 2018; Wen et al. 2025).

Vegetation restoration also enriched the soil–root system, which, in turn, promotes the formation of a layer of dead branches and fallen leaves, thereby increasing the accumulation of soil humus and leading to higher SOM content (Anning et al. 2023). In the present study, the organic matter contents in forestlands ($3.83\,g{\cdot}kg^{-1}$) and grasslands ($3.42\,g{\cdot}kg^{-1}$) were significantly higher than those in farmlands ($2.26\,g{\cdot}kg^{-1}$), which supports the view that organic matter enrichment facilitates aggregate formation and macropore development, thereby promoting preferential flow pathways (Hu et al. 2025). These processes enhance soil infiltration and impact vertical and lateral water redistribution.

4.2 | Controlling Effects of Land Use on Preferential Flow Contribution

One-way ANOVA revealed clear differences in preferential flow characteristics among different land-use types. Grasslands exhibited the highest P_{FR}, whereas forests had the highest LI. Nevertheless, both the P_{FR} and LI in forestlands exceeded those in farmlands. This indicates that vegetation restoration measures promote the development of preferential flow, which is consistent with the findings of Zhang et al. (2025). Additionally, the length indices for the forestland and grassland sites were comparable, indicating a relatively similar extent of preferential flow development. Among the three farmland sites, LI was nearly identical across all three farmland sites, reaching approximately 100%, suggesting that the potato and broomcorn millet farmlands selected for this experiment exhibited a comparable degree of preferential flow development.

The stained area in each soil profile decreased with depth, and both the maximum dyeing depth and coverage were highest in forestland, followed by grassland and farmland. This difference is primarily attributed to *Ks*, which was notably higher in forestland ($23.24\,\mathrm{mm{\cdot}h^{-1}}$), compared to grassland ($7.51\,\mathrm{mm{\cdot}h^{-1}}$) and farmland ($13.10\,\mathrm{mm{\cdot}h^{-1}}$), facilitating faster water transport to deep soil layers. However, the CPF to total infiltration was highest in grassland (46.79%), followed by forestland (21.50%) and farmland (23.05%) (Figure 6). Several mechanisms may explain the higher preferential flow contribution observed in grasslands. First, grassland root systems are typically dense, fibrous, and concentrated in the upper and middle soil layers. These roots tend to form numerous fine interconnected macropores and biopores, enhancing lateral connectivity and facilitating rapid bypass flow during infiltration events (Mei et al. 2018; Guan et al. 2025; Wen et al. 2025). In comparison, forestland roots are generally deeper and more heterogeneous, which may favor vertical preferential flow, but also increase matrix flow in deeper layers, thereby reducing the proportional CPF to total infiltration. Second, the surface conditions may also play an important role. Grasslands usually experience less mechanical disturbance than farmlands, while lacking the thick litter layers commonly found in forests. These intermediate conditions may allow near-surface preferential pathways to be activated more readily during infiltration. Previous studies have reported higher susceptibility to macropore flow in grasslands than in forests under similar climatic conditions (Alaoui 2015; Alaoui et al. 2011). Together, these factors explain why grasslands exhibited a higher proportional CPF, despite forestlands exhibiting greater infiltration depth and DC. This indicates that the preferential flow contribution depends not only on the extent of the stained areas or infiltration depth, but also on the efficiency and connectivity of the flow pathways.

4.3 | Contributions, Limitations, and Future Perspectives

Numerous studies have demonstrated that converting farmland into forest or grassland improves the physical and chemical properties of the soil and promotes preferential flow (Wen et al. 2025; Zhang et al. 2025). In our study, we explored the hypothetical causal relationships among different land-use types, basic soil physical and chemical properties, preferential flow characteristics, and preferential flow contributions using PLS-SEM. Land use exerted strong indirect effects on preferential flow indices through its effects on root biomass and soil physicochemical properties, particularly BD and porosity. The negative relationship between BD and LI highlights the critical role of soil structural loosening in enhancing preferential pathway continuity. The model also indicated that preferential flow morphology and contribution responded differently to vegetation restoration. While forestland strongly promoted preferential flow depth and length, grassland more effectively increased the relative CPF to the total infiltration. This divergence highlights the importance of distinguishing between preferential flow development and its hydrological effectiveness when evaluating vegetation restoration outcomes, which emphasizes the ecological and hydrological significance of the impact of vegetation restoration on preferential flow, as revealed in this study.

In addition, several limitations should be acknowledged in our study has several limitations. The relatively small sample size may constrain the statistical power; however, PLS-SEM is suitable for exploratory analysis under limited sample conditions when model complexity is controlled, and paths are theory-driven. In addition, measurements were conducted during a single period, and the seasonal dynamics of soil moisture, root activity, and preferential flow were not captured. Root biomass was assessed only in surface soils, which may not fully represent deeper root contributions, particularly in forestlands. Future studies should increase the sample size, incorporate seasonal observations, and include more detailed root system characterization to better resolve vegetation-soil-preferential flow interactions.

5 | Conclusion

This study examined how vegetation restoration affects soil preferential flow characteristics and its role in infiltration processes in the Loess Plateau. By comparing farmland, forestland, and grassland, we demonstrated that vegetation restoration enhanced the preferential flow development relative to farmland, whereas different vegetation types regulated infiltration through distinct preferential flow patterns. Differences in preferential flow characteristics between vegetation types were consistently associated with variations in soil physical properties, including porosity, BD, and root biomass. These associations suggest that vegetation-induced soil structural heterogeneity provides an important internal basis for preferential flow. Infiltration during vegetation restoration is primarily dominated by piston flow. Although forestland exhibited deeper and more continuous preferential pathways, grassland showed the highest CPF to the total infiltration. This contrast indicates that preferential flow acts as an internal regulator, which redistributes infiltration pathways and does not directly control the total infiltration. Such internal regulation highlights the importance of preferential flow contribution, rather than morphology, in shaping the infiltration behavior. Therefore, in practice, vegetation restoration strategies should consider not only improving the soil structure but also optimizing the preferential flow contribution to balance soil water retention and deep percolation in the Loess Plateau.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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